F172 Lesson 10: Gene Therapy Clinical Trials & Task 2 (ANT-130)

OCR AAQ Level 3 Human Biology · Unit F172: Genetics (Matt)

Clinical Trial Phases | ANT-130 Case Study | Biomarker Analysis | NEA Task 2 Criteria

Assessment Focus

Targeted Preparation: Complete preparation for OCR Genetics NEA Task 2.

Core Objective: Evaluating gene therapy research and clinical application.

Grading Alignment: Direct pathways to Pass, Merit, and Distinction criteria.

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Clinical Trial Architecture: From Bench to Bedside

Phase I (Safety & Dosage)	**Small Cohorts:** Tested in highly restricted groups (N=10–30 participants).	**Primary Goal:** Assesses acute safety and procedure tolerability.	**Toxicity Check:** Identifies adverse dose-limiting toxicities.	**Dose Escalation:** Determines the optimal and safe dosage range.
Phase II (Proof of Concept)	**Medium Cohorts:** Expanded participant testing groups (N=30–100).	**Target Engagement:** Evaluates if the drug hits the biological target.	**Biomarker Tracking:** Monitors surrogate biomarkers for early biological signals.	**Preliminary Efficacy:** Looks for initial signs of clinical benefit.
Phase III (Pivotal Trial)	**Large Cohorts:** Massive multi-centre cohorts (100s–1000s of patients).	**Gold Standard:** Double-blind randomized controlled trials (RCTs).	**Definitive Proof:** Proves clinical benefit versus standard of care.	**Sham/Placebo:** Compares directly to placebo or sham procedures.

Regulatory Milestones: MHRA (UK), EMA (Europe), and FDA (US) investigational approvals required.

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Huntington’s Disease: Molecular Etiology & Neurodegeneration

Genetic Mutation

- **Inheritance:** Autosomal dominant expansion of CAG trinucleotide repeats.
- **Threshold:** Pathological when sequence exceeds >36 CAG repeats.
- **Locus:** Located in exon 1 of the HTT (IT15) gene on chromosome 4.

Toxic Polyglutamine Protein

- **Translation:** Expanded CAG creates a massive polyglutamine domain.
- **Misfolding:** Results in toxic mutant huntingtin (mHTT) protein.
- **Aggregation:** Forms destructive intracellular clumps inside neurons.

Striatal Neurodegeneration

- **Cell Death:** Selective loss of GABAergic medium spiny neurons (MSNs).
- **Anatomy:** Primarily affects the corpus striatum (caudate nucleus and putamen).
- **Symptoms:** Triggers progressive motor chorea and cognitive decline.

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Case Study: ANT-130
Molecular Mechanism of Action

Vector Architecture

- **Delivery Vehicle:** Recombinant AAV5 viral capsid (adeno-associated virus).
- **Payload:** Carries an artificial, engineered microRNA (miRNA / miHTT).
- **Target:** Specifically designed to bind human HTT mRNA transcripts.

Stereotactic Delivery

- **Surgical Route:** Single, permanent intracranial neurosurgical microinfusion.
- **Technology:** Uses MRI-guided Convection-Enhanced Delivery (CED).
- **Target Zones:** Bilateral injection directly into the caudate and putamen.

RNA Interference (RNAi)

- **Processing:** Intracellular miRNA processing by the Dicer enzyme.
- **Activation:** Loaded into the RNA-induced silencing complex (RISC).
- **Gene Silencing:** Cleaves HTT mRNA, permanently lowering toxic mHTT protein.

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ANT-130 Phase I/II Clinical Trial Design (NCT04120493)

Cohort Architecture

- **Trial Structure:** Multi-centre, randomized, double-blind design (29 patients).
- **Dosing Arms:** Includes distinct low-dose (N=12) and high-dose (N=17) cohorts.
- **Control Arm:** Utilizes an imitation (sham) surgical control group.

Sham Surgical Control

- **Imitation Surgery:** Involves superficial cranial bone drilling only.
- **Safety Precaution:** No actual dural penetration or brain entry occurs.
- **Scientific Validity:** Ethically controls for placebo effects and surgical expectations.

Longitudinal Duration

- **Extended Timeline:** Multi-year patient follow-up (up to 36–60 months).
- **Vector Tracking:** Assesses persistent AAV5 vector expression in the brain.
- **Clinical Tracking:** Monitors long-term neurological and motor stability.

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ANT-130 Biomarker Findings: CSF mHTT & NfL Dynamics

CSF mHTT Lowering

Target Engagement: Confirms the miRNA therapy successfully hit its molecular target.

Significant Drop: Statistically significant reduction of mutant huntingtin (mHTT) protein.

Dose-Dependent: Greater mHTT reductions observed in high-dose cohorts.

Sustained Effect: Lowered levels maintained long-term in cerebrospinal fluid (CSF).

Neurofilament Light Chain (NfL)

Trauma Marker: NfL is a well-characterized biomarker of active neurodegeneration.

Surgical Spike: Acute, transient NfL elevation immediately post-procedure from surgery.

Long-Term Decline: Sustained -8.2% drop below baseline observed at 36 months.

Neuroprotection: Indicates definitively decelerated active neuroaxonal loss.

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Clinical Functional Efficacy: Slowing Disease Progression

Composite UHDRS (cUHDRS)

- **Holistic Measure:** Tracks composite cognitive, motor, and functional capacity.
- **High-Dose Success:** High-dose cohort showed significant functional preservation.
- **Disease Slowing:** Demonstrated a statistically significant 75% slowing of progression ($p=0.003$) at 36 months.
- **Baseline Comparison:** Measured against propensity score-matched natural history data.

Total Motor & Functional Scores

- **Total Motor Score (TMS):** Quantifies chorea and rigidity; showed a 59% slowing of progression.
- **Total Functional Capacity (TFC):** Key secondary endpoint met; showed a 60% slowing of progression.
- **Sustained Benefit:** Efficacy maintained across 24 and 36-month clinical checkpoints.
- **Dose Response:** Stronger physical preservation observed in higher-dose patients.

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Evaluating Gene Therapy: Challenges & Delivery Risks

Surgical Delivery Risks

- **Invasive Nature:** CED requires highly invasive, open neurosurgery.
- **Infection Risk:** Carries inherent risks of intracranial hemorrhage or infection.
- **Precision Demanded:** Requires exact MRI-guided targeting of the striatum.

Irreversibility

- **Permanent Alteration:** AAV5 vector delivery is a permanent genetic alteration.
- **No Undo Button:** The therapeutic vector cannot be removed once administered.
- **Long-term Unknowns:** Lifelong effects of continuous miRNA expression are unknown.

Immune Responses

- **Viral Antibodies:** Pre-existing AAV5 neutralizing antibodies could block the vector.
- **Neuroinflammation:** Risk of the immune system attacking the modified brain cells.

Off-Target Effects

- **Unintended Silencing:** Risk of miRNA binding to non-HTT mRNA transcripts.
- **Wild-Type Impact:** Lowers both toxic mutant AND healthy (wild-type) huntingtin protein.

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Patient Support: The Role of Genetic Counselling

Pre-Symptomatic Complexity

HD has 100% penetrance but adult-onset. Patients can test positive decades before symptoms appear, creating an immense psychological and existential burden.

Informed Decision Making

- **Testing Choices:** Assisting families in deciding whether to undergo predictive testing.
- **Reproductive Options:** Discussing IVF and pre-implantation genetic diagnosis.

Trial Consent & Ethics

- **Managing Expectations:** Ensuring patients understand the lack of guaranteed cures.
- **Sham Surgery Consent:** Preparing patients for the possibility of imitation surgery.

Psychological Support

- **Result Processing:** Helping patients process positive HTT mutation results.
- **Family Dynamics:** Navigating the guilt, anxiety, and impact on untested relatives.

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NEA Task 2: Genetics Coursework Alignment Matrix

Pass Criteria (Foundational Knowledge)		
• **P3 & P4 (Pathology/Inheritance):** Autosomal dominant, CAG repeats >36, chromosome 4 (Slide 3).	• **P6 & P7 (Mechanism/Delivery):** AAV5 vector, engineered miRNA, RNAi silencing, CED neurosurgery (Slide 4).	• **P8 (Counselling):** Summarize the role of the genetic counsellor (Slide 9).
Merit Criteria (Analysis & Expression)		
• **M2 (Gene Expression):** Expanded polyglutamine tract causing toxic mHTT aggregation (Slide 3).	• **M3 & M4 (Risks & Challenges):** Analyze surgical risks, irreversible nature, and immune response (Slide 8).	• **M5 (Counselling Benefits):** Explain psychological support for adult-onset diseases (Slide 9).
Distinction Criteria (Evaluation)		
• **D1 (Physiological Impact):** Loss of GABAergic MSNs causing chorea and cognitive decline (Slides 3 & 7).	• **D2 (Advantages/Disadvantages):** Contrast clinical efficacy (cUHDRS) vs permanent surgical risks (Slides 7 & 8).	• **D3 (Therapy Relevance):** Assess why AAV5-miHTT is the most viable path to a functional cure for HD.

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